

John Gerega

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Technical Summary

Computer Science graduate (GPA: 3.91, Summa Cum Laude) with strong foundations in software development, data structures, backend systems, and scalable application design. Experienced in building full-stack applications using Python, Java, JavaScript, and C, with hands-on work in Flask, React, SQL Server, and Oracle databases. Strong interest in AI/ML systems, APIs, and data-driven backend infrastructure.

Experience & Projects

Full Stack Developer Intern — Encompass (Remote) • May 2025 – August 2025

- Developed backend services and REST APIs for a HIPAA-compliant patient dashboard ensuring secure data access
- Designed relational database schema with structured data models and efficient query patterns
- Implemented authentication flows and protected routing for secure user session management
- Collaborated using GitHub and Azure DevOps in an agile SDLC environment

Software Engineering Intern — Encompass (Remote) • July 2024

- Developed a client-server file transfer system using socket-based communication
- Implemented encryption techniques to ensure secure transmission and data integrity
- Optimized system performance through iterative testing and architectural improvements

Vulcan Activity Tracker (Senior Capstone Project) • Aug 2025 – May 2026

- Architected a modular Flask backend using a multi-blueprint structure for separation of concerns
- Designed normalized SQL Server schema with constraints, validation logic, and efficient query design
- Implemented RESTful APIs supporting data ingestion, aggregation, and leaderboard ranking
- Developed dynamic frontend components and integrated real-time data visualization
- Built secure authentication using password hashing and session-based management
- Containerized full application stack using Docker Compose
- Utilized AI-assisted development tools to troubleshoot complex compiler and backend issues, accelerate debugging workflows, and improve implementation efficiency

Peer Tutor – Computer Science • Pennsylvania Western University • Aug 2023 – May 2026

- Assist students with debugging, algorithm design, and understanding time complexity
- Guide structured problem-solving approaches and efficient coding practices
- Applied concepts from Language Translation coursework including scanners, parsers, file readers, and code generation while working in C

Technical Skills & Experience

Languages	Python, Java, C, C++, C#, JavaScript, SQL, COBOL, Assembly
Web	Flask, React, HTML, CSS, REST APIs
Databases	SQL Server, MySQL, Oracle, Relational Database Design
Tools	Git, Docker, Azure DevOps, JIRA, VS Code
Concepts	Data Structures, Algorithms, OOP, Operating Systems, SDLC, Compiler Design, Parsing, Code Generation

Education

Pennsylvania Western University

B.S. Computer Science — Graduated May 2026

GPA: 3.91 • Summa Cum Laude Honors

Awards & Recognition

- Arthur and Alillicent Gabriel Merit Award for Outstanding Achievement in Computer Science and Computer Information Systems Students

Leadership, Athletics & Activities

- Vice President, Club Soccer — Pennsylvania Western University: Coordinated intercollegiate matches by organizing schedules and communication across multiple university teams; managed team logistics, scheduling, and internal communication workflows; maintained team engagement through social media outreach; provided leadership to ensure consistent participation and organization
- Member, Software Development Club — Participated in technical workshops and projects
- Arsenal FC Soccer (5 years) — Scholarship Recipient: Selected for scholarship recognizing leadership, discipline, and team-first mentality; consistently arrived early to training, maintained high attendance, and mentored younger players during sessions
- Pittsburgh Marathon Finisher (2x): Completed the Pittsburgh Marathon twice, improving from a 4:05 finish to a sub-4-hour 3:59 marathon